

tutions, and want returns on their loans. While it seems cold to think of it this way, the fact is, a bank does not want to lend money to a farmer whose crop will fail and then he/she will be unable to payback the loan. If the loan is large and the farmer commits suicide, there is often no collateral that the bank can use to recover their losses. What this results in is banks being very conservative with loans to farmers, which only makes the situation even worse.

This is where data collection of soil health is important, and a few startups are getting into this field. Working with the banks and farmers for a small commission, a startup can test and maintain records of farmlands across very large areas (entire states or the entire country if need be). The more frequent the data collection / updation, the more accurate the data, and thus, the more reliable the outcome prediction. Tied into a more accurate weather prediction model, and banks will be able to more accurately gauge the risk of their loan, and will be more willing to loosen their purse strings. This is also a win for the farmers, as having more money means being able to plant more expensive crops or just increase the yield of existing crops, and also being able to afford emergency solutions, such as buying tanker water or paying for irrigation in case the monsoon is delayed.

Big data is going to rule the Indian farming industry, and data on soil health is very much going to be a primary level focus for the future.

CROPS AND SEEDS

We've all heard of the hate that Indian farmers have for the large corporations that produce genetically modified (GM) seeds that may grow far better in adverse conditions, but also cost farmers a lot. Then there's the problem of not being able to use seeds from the crops you harvest to just plant more crops. It's like a subscription service that you have to keep paying for, and that's not something most Indian farmers can afford.

This is where a startup like Desi Seed Producers Company Ltd come in. While not the most creative name,

the company was set up with one of the most noble goals for Indian farming. Based in Bengaluru, the company sells the seeds under the brand name Sahaja Seeds, which are basically organically grown, patent-free and open pollinated seeds that are in the public domain when it comes to licensing. What this means is that a farmer buys only the seeds, and is then free to plant and harvest for as many cycles as he or she wants or needs.

While we really have nothing against GM seeds, because they are important in their own right, it's important that options like Sahaja Seeds exist, in order to keep the GM manufacturers honest, and also to reduce the prices.



Image: agrariancrisis.in

Syed Ghani Khan is a rockstar in the Indian organic farming community. He's single-handedly conserving over 550 varieties of rice, 140+ rare types of mango trees, 36 types of millet and even has a seed museum in his home

Another aspect of farming is crop rotation and also looking after the crops that are growing. There are some really innovative solutions that are coming out of India in this regard. There are startups using drones to take thermal imaging of crops, which helps them identify problematic areas in a large field, and thus adjust the irrigation or nutrition given to those patches. It's also a good way to identify pests, by identifying dying plants in a large plantation. Many others are also using AI and image recognition to allow a farmer to click a picture of a plant that's not flourishing, and a system will diagnose it with possible problems and next steps. This way, farmers can know how to deal with and rescue a dying patch or farm.

Another aspect that's important in India is water usage, and preventing water wastage. Given that the majority of crop failures are due to drought or flood, it's obvious that although water is essential to farming, it's also the bane of their existence. Various techniques and ideas are being used to aid in water conservation, which include drip irrigation, mulching, the setting up of automatic sprinkler systems, and more. The pervasiveness of technology means that keeping an eye on crops and preventing losses is a lot easier now and can be achieved with far more efficiency than ever before. The Internet of Things (IoT) is going to play a big role in this area in the coming decades – more and more

places across the country are getting access to cheap wireless (cellular) internet, which is a requirement for IoT.

EQUIPMENT

Farming equipment is expensive, and most small farmers have been forced to use livestock, or even human power to till their lands and harvest their crops. This works on really small farms, but when you have dozens of acres to till, maintain and harvest, you're going to need machinery to assist you. With small profit margins and high expenses, it's no wonder that many farmers cannot afford to buy farming equipment. This is where startups who are providing rental services to farmers for lower costs come in. Instead of investing lakhs on buy-